

least on the downside, and at least when the volatility could have the largest impact.

Spending can be kept constant if the portfolio is derisked. To really get constant spending, you should look to hold fixed-income assets to maturity or use risk-pooling assets like income annuities. Consider, for instance, traditional defined-benefit pension systems that can reduce volatility's impact on the pensioner's income. Defined-benefit pensions provide a way to pool the sequence-of-returns risk across separate birth cohorts, thus reducing exposure. Individuals are entitled to benefits based on their contributions into the system, not market performance. With defined-benefit pensions, some individuals will receive less than they might have by investing on their own, but others will receive more. In this regard, defined-benefit pensions are essentially a separate asset class that most investors should find very valuable, as pensions diversify sequence of returns across time and allow them to collect income based more closely on the average returns over long periods.

Other approaches to reducing downside risk (volatility in the undesired direction) could also be considered. For instance, a rising equity glide path in retirement could start with an equity allocation that is even lower than typically recommended (in safe withdrawal rate research literature) at the start of retirement, but then slowly increase the stock allocation over time. This can reduce the probability and the magnitude of retirement failures. This approach reduces vulnerability to early retirement stock market declines that cause the most harm to retirees. Asset allocation could also be managed with a funded ratio approach, in which more aggressive asset allocations are used only when sufficient assets are available beyond what is necessary to meet retirement spending goals. Finally, financial derivatives or income guarantee riders can be used to place a floor on how low a portfolio may fall by sacrificing some potential upside.

4. Buffer Assets—Avoid Selling at Losses

The final category of approaches to managing sequence risk is to have other assets available outside the financial portfolio to draw from after a market downturn. Returns on these assets should not be correlated with the